

Causal Inference Over Time: Developing Methods for Longitudinal Data with Applications in Health Research

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Introduction

Motivation

- ▶ Inform decision-makers about the real-world value of statin use for primary prevention of cardiovascular disease (CVD) in older adults (aged 65+) in Quebec
- ▶ Three-part project :
 - ▶ Methodological development
 - ▶ Epidemiological analysis of the cohort
 - ▶ Cost-effectiveness evaluation
- ▶ This presentation contributes to the first component : development of causal inference methods

Background and Problem

- ▶ Cardiovascular diseases (CVDs) are the leading cause of death worldwide
- ▶ They are the second leading cause of death in Canada
- ▶ Older adults are particularly at risk
- ▶ Identifying effective prevention strategies is essential for healthy aging and healthcare sustainability



Background and Problem

- ▶ Statins prevent CVD by lowering cholesterol
- ▶ Their effectiveness is well established in the general population[Taylor et al, 2011].
- ▶ Evidence in adults aged 65+ is limited and inconclusive [Singh et al, 2018].
- ▶ Administrative data provide an opportunity to study these questions at scale



Notation

We observe :

$$O = (X_0, A_0, X_1, A_1, C_1, \dots, X_T, A_T, C_T, Y) \quad (1)$$

- ▶ A_t : binary treatment at time t
- ▶ Y : outcome at the end of follow-up
- ▶ X_t : vector of covariates at time t
- ▶ C_t : censoring indicator at time t
- ▶ $\bar{A}_t = \{A_0, \dots, A_t\}$: treatment history
- ▶ $\bar{X}_t = \{X_0, \dots, X_t\}$: covariate history
- ▶ $Y^{\bar{a}_T}$: potential outcome under treatment regime $\bar{a}_T$

Challenges in Administrative Data

- **Main issue** : time-dependent confounding

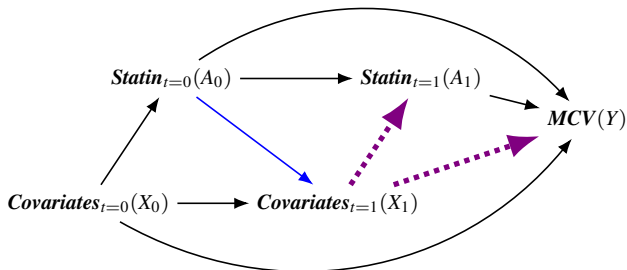


Figure : DAG (Directed Acyclic Graph) illustrating the dual role of time-varying covariates with two time points.

- **Solution** : Marginal Structural Models (MSM).

Marginal Structural Models (MSM)

Causal effect estimation

$$\mathbb{E}(Y^{\bar{a}_T}) = \beta_0 + \beta_1 \cdot \text{cum}(\bar{a}_T) \quad (2)$$

$$\text{cum}(\bar{a}_T) = \sum_{t=0}^T a_t$$

- β_1 : effect of one additional month of exposure

Marginal Structural Models (MSM)

Causal effect estimation

$$\mathbb{E}(Y^{\bar{a}_T}) = \beta_0 + \beta_1 \cdot \text{cum}(\bar{a}_T) \quad (2)$$

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- ▶ β_1 : effect of one additional month of exposure

Key assumptions

- ▶ Sequential exchangeability :

$$Y^{\bar{a}_T} \perp\!\!\!\perp A_t \mid \bar{A}_{t-1}, \bar{X}_t$$

- ▶ Positivity :

$$0 < P(A_t = 1 \mid \bar{A}_{t-1}, \bar{X}_t) < 1$$

- ▶ No interference : one version of treatment

- ▶ Consistency : if $\bar{A}_T = \bar{a}_T$, then $Y = Y^{\bar{a}_T}$

Inverse Probability of Treatment Weighting (IPTW)

Step 1 : Create a pseudo-population by weighting each individual using stabilized (3) or marginal stabilized weights (4)

$$U_{a(t)} = \prod_{k=0}^t \frac{P(A_k = a_{ik} | \bar{A}_{k-1} = \bar{a}_{i,k-1})}{P(A_k = a_i | \bar{A}_{k-1} = \bar{a}_{i,k-1}, \bar{X}_k = \bar{x}_{i,k})} \quad (3)$$

$$W_{a(t)} = \prod_{k=0}^t \frac{P(A_k = a_{ik})}{P(A_k = a_i | \bar{A}_{k-1} = \bar{a}_{i,k-1}, \bar{X}_k = \bar{x}_{i,k})} \quad (4)$$

Step 2 : Fit a weighted regression model using these weights to estimate the causal effect

Inverse Probability of Censoring Weighting (IPCW)

Step 1 : Account for censoring by weighting each individual

Stabilized censoring weights :

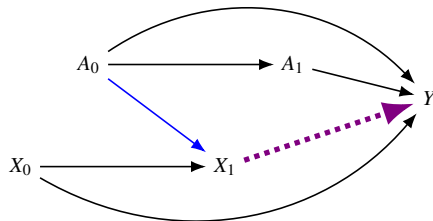
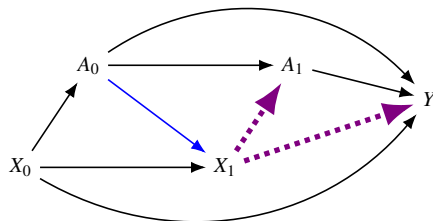
$$V_{c(t)} = \prod_{k=0}^t \frac{P(C_k = 0 \mid \bar{C}_{k-1} = 0)}{P(C_k = 0 \mid \bar{C}_{k-1} = 0, \bar{A}_{k-1} = \bar{a}_{k-1}, \bar{X}_k = \bar{x}_k)} \quad (5)$$

Step 2 : Combine with treatment weights and fit a weighted model

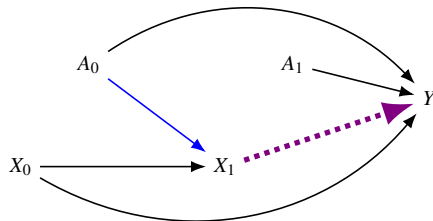
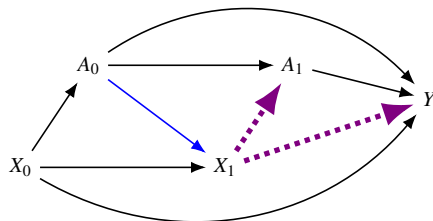
Final weight : $W_{a(t)} \times V_{c(t)}$

to estimate the causal effect in the presence of censoring

Stabilized Weights



Marginal Stabilized Weights



Checking Covariate Balance

1. Using $U_{a(t)}$:

$$A_t \perp\!\!\!\perp \bar{X}_{t-1} \mid \bar{A}_{t-1}$$

for $t = 1, \dots, T$ and any treatment history $\bar{A}_{t-1}$.

2. Using $W_{a(t)}$:

$$A_t \perp\!\!\!\perp \bar{X}_{t-1}$$

for $t = 1, \dots, T$.

Goal : Assess whether treatment is independent of covariates after weighting

Literature Review

- Several balance metrics have been proposed [Franklin, 2014]

Single-variable metrics :

1. Absolute difference (D),
2. Standardized mean difference (SMD),
3. Overlap coefficient (OVL),
4. Kolmogorov–Smirnov distance (KS),
5. Lévy distance (LD).

Literature Review

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Multivariate metrics :

1. Post-weighting C-statistic (CS),
2. General weighted difference (GWD),
3. Mahalanobis balance (MHB).

Literature Review

Standard balance metrics

- Absolute difference

$$D = \hat{\mathbb{E}}[X|A = 1] - \hat{\mathbb{E}}[X|A = 0] \quad (6)$$

- Standardized mean difference

$$SMD = \frac{\hat{\mathbb{E}}[X|A = 1] - \hat{\mathbb{E}}[X|A = 0]}{\sqrt{\frac{\sigma_{A=1}^2 + \sigma_{A=0}^2}{2}}} \quad (7)$$

Literature Review

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Longitudinal extension

$$\hat{\mathbb{E}}[U_{1(t)}I(A_t = 1)X_{t-k}|\bar{A}_{t-1}] - \hat{\mathbb{E}}[U_{0(t)}I(A_t = 0)X_{t-k}|\bar{A}_{t-1}] \quad (8)$$

for $0 \leq k \leq t$

Literature Review

Example : Two time points ($T = 1$)

At $t = 0$:

$$\hat{\mathbb{E}}[X_0|A_0 = 1] - \hat{\mathbb{E}}[X_0|A_0 = 0]$$

At $t = 1$, for $A_0 \in \{0, 1\}$:

$$\hat{\mathbb{E}}[X_0|A_1 = 1, A_0] - \hat{\mathbb{E}}[X_0|A_1 = 0, A_0]$$

$$\hat{\mathbb{E}}[X_1|A_1 = 1, A_0] - \hat{\mathbb{E}}[X_1|A_1 = 0, A_0]$$

Literature Review

Example : Three time points ($T = 2$)

- At $t = 0$:

$$\hat{\mathbb{E}}[X_0|A_0 = 1] - \hat{\mathbb{E}}[X_0|A_0 = 0]$$

- At $t = 1$, for $A_0 \in \{0, 1\}$:

$$\hat{\mathbb{E}}[X_0|A_1 = 1, A_0] - \hat{\mathbb{E}}[X_0|A_1 = 0, A_0]$$

$$\hat{\mathbb{E}}[X_1|A_1 = 1, A_0] - \hat{\mathbb{E}}[X_1|A_1 = 0, A_0]$$

- At $t = 2$, for (A_0, A_1) :

$$\hat{\mathbb{E}}[X_j|A_2 = 1, A_0, A_1] - \hat{\mathbb{E}}[X_j|A_2 = 0, A_0, A_1], \quad j = 0, 1, 2$$

⇒ **Total : 17 balance conditions**

Key Challenges

- ▶ Number of diagnostics grows rapidly
- ▶ No clear threshold for imbalance
- ▶ No consensus on best metric

Objective

Main objective

To evaluate covariate balance metrics in longitudinal settings with time-varying treatments

Specifically

- ▶ Compare existing balance metrics
- ▶ Assess their ability to detect imbalance
- ▶ Evaluate their relationship with estimation bias

Proposed Approach

General strategy

- ▶ Compare groups $A_t = 0$ vs $A_t = 1$
- ▶ Use weighted data $W_{a(t)}$
- ▶ Threshold : $p \times 0.01$ for MHB

Simulation study

- ▶ 10 scenarios (7 from Franklin et al. + 3 new)
- ▶ Two time points : treatments (A_0, A_1) , covariates (X_0, X_1) , binary outcome Y
- ▶ Bias modeled as a function of each metric (linear + quadratic)
- ▶ Performance criteria : high R^2 and intercept (β_0) close to zero

Simulation Results

Key message : Mahalanobis-based metric (MHB) best reflects bias

Tableau : Intercept (β_0) and explained variance (R^2) across 10 simulation scenarios

	Scenario 1		Scenario 2		Scenario 3		Scenario 4		Scenario 5	
	R^2	β_0	R^2	β_0	R^2	β_0	R^2	β_0	R^2	β_0
D	0.94	-0.09	0.95	-0.05	0.71	-0.26	0.01	0.04	0.76	0.01
SMD	0.95	-0.08	0.96	-0.04	0.74	-0.20	0.02	-0.05	0.77	0.00
OVL	0.84	-4.19	0.92	-0.04	0.35	-3.04	0.09	-1.71	0.67	7.94
KS	0.91	-0.31	0.95	-0.23	0.61	-0.73	0.09	-0.21	0.76	-0.02
LD	0.90	-0.30	0.95	-0.20	0.61	-0.72	0.07	-0.15	0.76	-0.02
MHB	0.97	0.02	0.97	0.00	0.77	0.06	0.02	-0.16	0.76	0.01
CS	0.49	-0.28	0.51	-0.32	0.48	-0.27	0.03	0.10	0.53	-0.03
GWD	0.94	-0.14	0.96	-0.03	0.70	-0.43	0.01	-0.05	0.77	0.00
	Scenario 6		Scenario 7		Scenario 8		Scenario 9		Scenario 10	
	R^2	β_0	R^2	β_0	R^2	β_0	R^2	β_0	R^2	β_0
D	0.88	-0.58	0.91	-0.10	0.97	-0.02	0.91	-0.08	0.97	-0.02
SMD	0.93	-0.35	0.92	-0.07	0.97	-0.01	0.92	-0.03	0.97	0.00
OVL	0.60	-0.36	0.77	-2.66	0.95	1.55	0.69	17.49	0.94	1.52
KS	0.84	-0.69	0.88	-0.21	0.97	-0.02	0.90	-0.24	0.97	-0.01
LD	0.84	-0.70	0.88	-0.19	0.97	-0.02	0.90	-0.16	0.97	-0.01
MHB	0.95	-0.17	0.93	0.01	0.97	-0.02	0.93	-0.03	0.97	-0.01
CS	0.52	-0.59	0.53	-0.16	0.51	-0.11	0.61	-0.31	0.51	-0.11
GWD	0.91	-0.42	0.89	-0.17	0.97	-0.01	0.91	-0.09	0.97	-0.01

Scenarios : 1 : baseline, 2 : rare exposure, 3 : small sample, 4 : high imbalance, 5 : low imbalance, 6 : strong confounding, 7 : nonlinear outcome, 8 : nonlinear exposure, 9 : redundant covariates, 10 : instrumental variables.

Application

- ▶ Retrospective cohort from Quebec administrative health databases.
- ▶ **Inclusion** : New statin users (2013–2018), age ≥ 66 , covered by public drug insurance.
- ▶ Exposure : Monthly statin use (binary).
- ▶ Outcome : First CVD event or all-cause death.
- ▶ Covariates : Age, diabetes, hypertension, treatment, kidney disease, hospitalizations.

Application

- ▶ 32,690 Quebec residents aged ≥ 66 , new statin users.
- ▶ Objective : Estimate the effect of statin use over time on time to first CVD event or death.
- ▶ Balance assessment :
 - ▶ Global : Mahalanobis balance (MHB)
 - ▶ Detailed : SMD if imbalance detected
- ▶ Analysis : Cox Marginal Structural Model (MSM)

$$\lambda_{\bar{a}_T}(\tau) = \lambda_0(\tau) \exp \left(\beta \sum_{t=0}^{11} a_t + \beta_1 V \right) \quad (9)$$

- ▶ Estimation using IPTW

Application Results

Comparison between statin users and non-users, before and after weighting IPTW $\times$ IPCW

		$A_3 = 1$ vs $A_3 = 0$		$A_4 = 1$ vs $A_4 = 0$		$A_5 = 1$ vs $A_5 = 0$	
		Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Covariables fixes dans le temps		0.06	0.07	0.06	0.10	0.05	0.11
Covariables variables dans le temps							
	0	0.04	0.07	0.04	0.04	0.04	0.01
	1	0.07	0.06	0.06	0.03	0.06	0.02
	2	0.12	0.05	0.09	0.03	0.08	0.02
Temps de mesure	3	0.17	0.03	0.13	0.02	0.11	0.02
	4			0.15	0.02	0.13	0.02
	5					0.15	0.02
		$A_6 = 1$ vs $A_6 = 0$		$A_7 = 1$ vs $A_7 = 0$		$A_8 = 1$ vs $A_8 = 0$	
		Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Time-invariant covariates		0.05	0.05	0.05	0.03	0.05	0.05
Time-varying covariates							
	0	0.04	0.01	0.04	0.06	0.04	0.01
	1	0.06	0.01	0.06	0.07	0.05	0.02
	2	0.08	0.01	0.08	0.07	0.07	0.02
	3	0.10	0.01	0.10	0.08	0.09	0.01
Time point	4	0.11	0.02	0.11	0.07	0.10	0.01
	5	0.13	0.02	0.12	0.07	0.12	0.02
	6	0.15	0.03	0.14	0.05	0.13	0.02
	7			0.15	0.08	0.14	0.01
	8					0.15	0.01

Time-invariant covariates : sex, age, region. Time-varying covariates : diabetes, hypertension, kidney disease, treatments, prescriptions, hospitalizations.

Application Results

Covariate balance over time (MHB)

Comparison between statin users and non-users, before and after weighting IPTW $\times$ IPCW

	$A_9 = 1$ vs $A_9 = 0$		$A_{10} = 1$ vs $A_{10} = 0$		$A_{11} = 1$ vs $A_{11} = 0$	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Time-invariant covariates	0.05	0.05	0.05	0.03	0.05	0.05
Time-varying covariates						
0	0.04	0.04	0.04	0.02	0.04	0.02
1	0.05	0.04	0.05	0.02	0.05	0.02
2	0.07	0.04	0.07	0.02	0.07	0.02
3	0.09	0.04	0.09	0.02	0.08	0.02
4	0.10	0.04	0.10	0.03	0.09	0.02
5	0.11	0.05	0.11	0.03	0.10	0.02
Time point						
6	0.12	0.05	0.11	0.03	0.11	0.02
7	0.13	0.04	0.12	0.02	0.11	0.02
8	0.14	0.04	0.13	0.02	0.12	0.02
9	0.15	0.03	0.14	0.02	0.13	0.03
10			0.16	0.02	0.14	0.04
11					0.16	0.02

Time-invariant covariates : sex, age, region. Time-varying covariates : diabetes, hypertension, kidney disease, treatments, prescriptions, hospitalizations.

Each additional month of statin use during the first year was associated with a 3% reduction in risk $HR = 0.97$, $IC95\% : 0.96 - 0.98$).

Conclusion

Key findings

- ▶ Stabilized weights reduce computational complexity
- ▶ Mahalanobis distance performs best for detecting imbalance
- ▶ Threshold : $p \times 0.01$
- ▶ Sustained statin use is associated with reduced risk of CVD or death

Conclusion

Key findings

- ▶ Stabilized weights reduce computational complexity
- ▶ Mahalanobis distance performs best for detecting imbalance
- ▶ Threshold : $p \times 0.01$
- ▶ Sustained statin use is associated with reduced risk of CVD or death

Limitations

- ▶ All covariates included in weight estimation
- ▶ Multiple valid adjustment sets may exist
- ▶ Need to balance bias reduction and variance

Merci pour votre attention !

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